LAST NAME: _____

Practice Quiz 10B — MATH 2055 — Fall 2026

Practice for Quiz 10

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

1. Prove by strong induction that every integer $n \geq 4$ can be written

$$n = 2a + 5b$$

for some nonnegative integers a, b. State the complete inductive hypothesis and verify every required base case.